

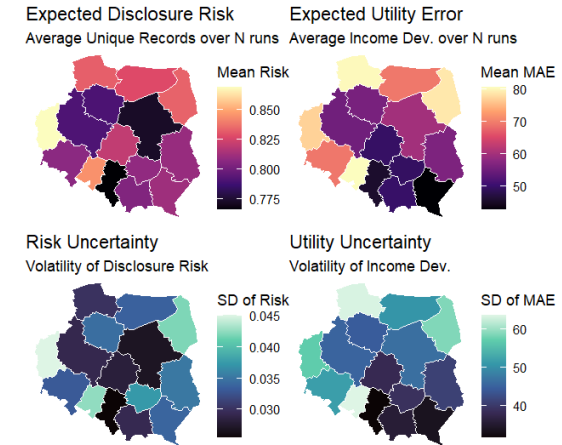
Balancing Privacy and Utility in Synthetic Data with Geospatial Dimensions

Violeta Calian and Nidia Guadalupe Lopez Flores
Statistics Iceland

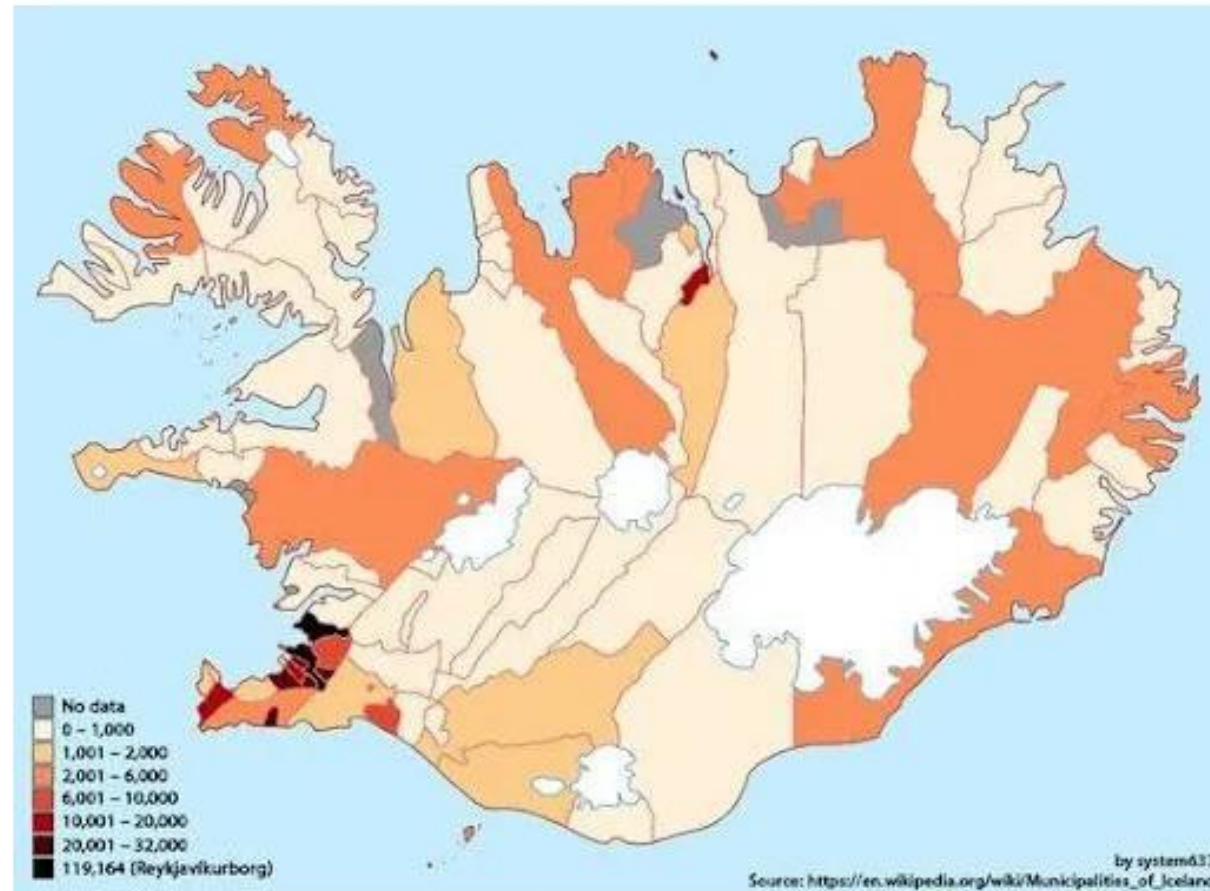
Content

- Formulation of the problem
- A method of solution: synthetic data
- Calibrating any synthesiser for risk-utility performance
- The impact of geospatial dimensions on SDC
- Confidence in tuning data protection on a map (w/wo DP)

Geographic Uncertainty in Synthetic Data Generation



Formulation of the problem: *why SDC on a map*



Multi-objective optimisation of synthesisers: risk-utility measures and method-hyperparameters

➤ Methods / with [R](#)

- sequential trees (CART or random forest) / [synthpop](#))
- joint probabilities
(Bayesian mixture models / [synMicrodata](#) or Bayesian Networks / [bnlearn](#))
- deep learning (adversarial NN / [RGAN](#))

➤ Measures (with **C.I.**) of:

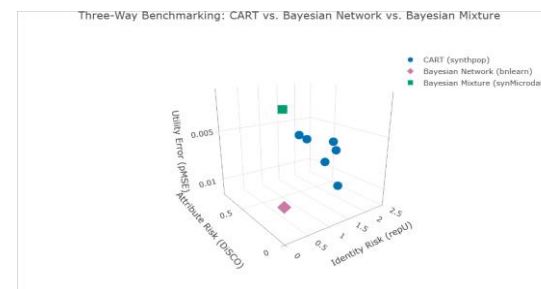
- Risk: *UiO*, *UiS*, *UiOiS*, *repU*, *DiSCO*, *Dorig*
- Utility: *pMSE*, *SPECKS*, *PO50*

➤ Problem:

- Multi-objective / Pareto optimization

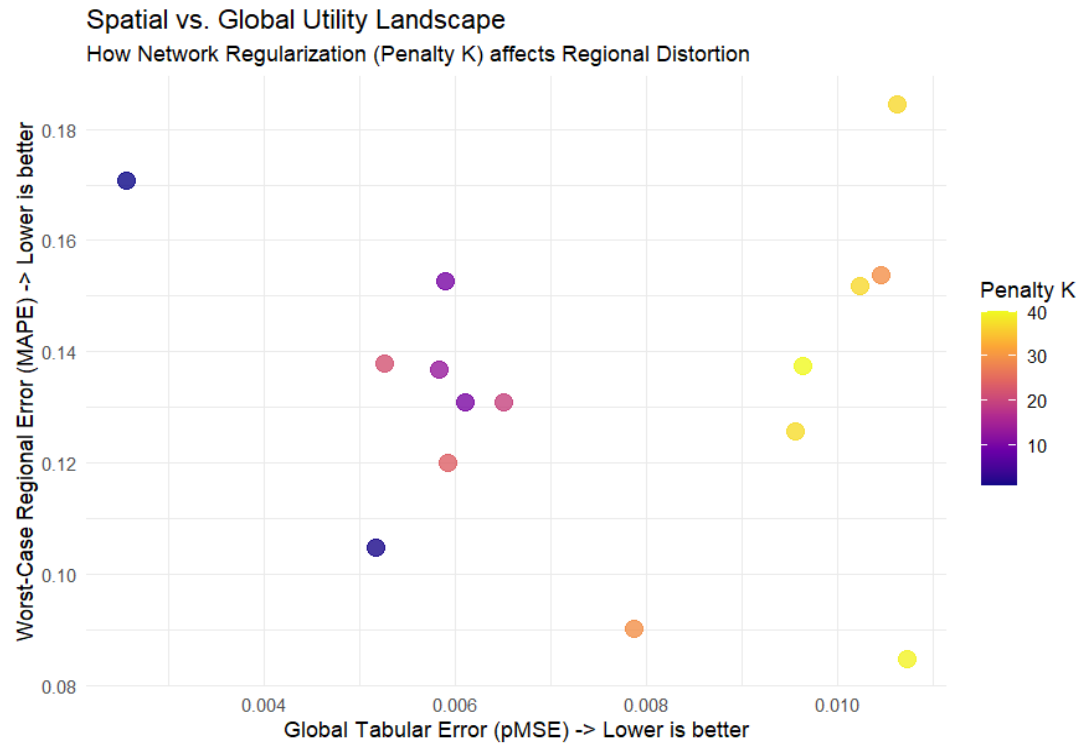
➤ Data for test/development: [open data](#): *SD2011* (Poland, [synthpop](#))

pMSE	SPECKS	PO50	repU	UiO	UiS	UiOiS	Dorig	DiSCO	max	Summary
0.004222146	0.2232	11.07	2.30	8.6	10.10	4.38	9.16	1.00		
0.011366370	0.1772	8.03	2.34	8.6	10.10	4.42	9.16	0.38		
0.007419048	0.1268	6.02	2.30	8.6	9.96	4.52	9.16	0.40		
0.014034703	0.2014	9.28	2.28	8.6	10.08	4.40	9.16	0.40		
0.004222146	0.2232	11.07	2.30	8.6	10.10	4.38	9.16	1.00		
0.011366370	0.1772	8.03	2.34	8.6	10.10	4.42	9.16	0.38		

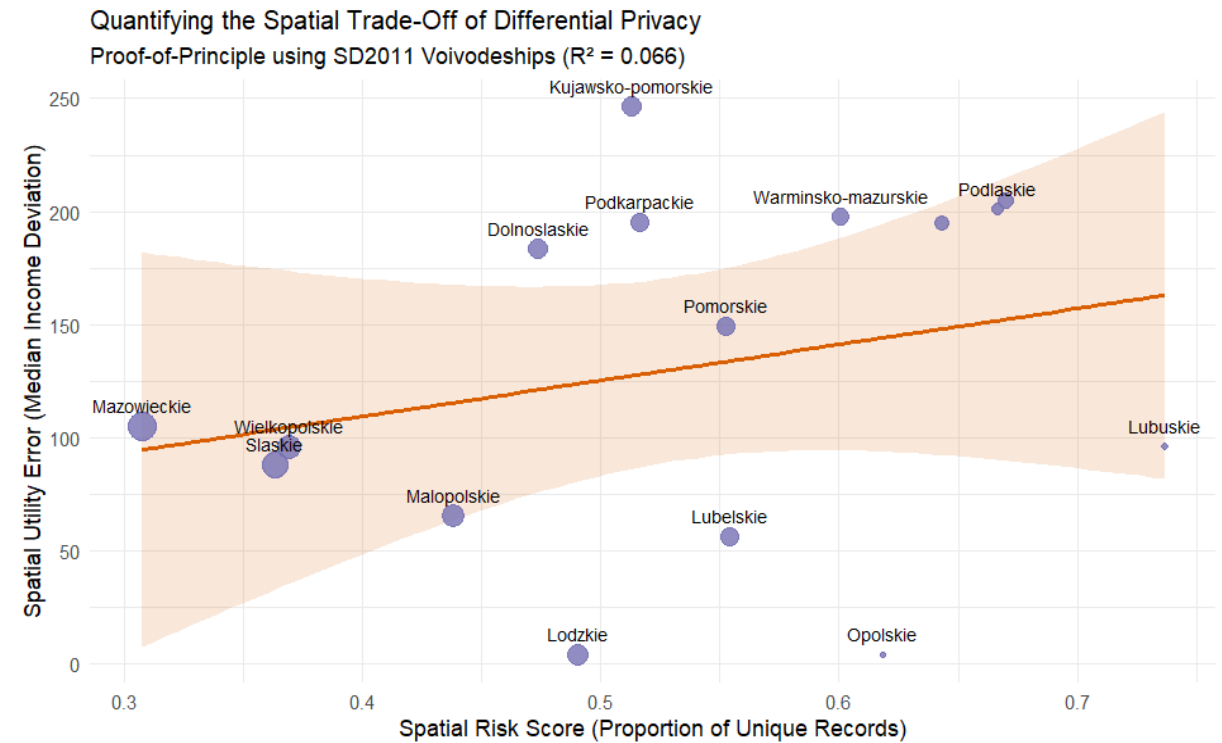


The impact of geospatial dimensions on SDC: *spatial sensitivity* (*why and „where?“*)

Bayesian Network synthetic data



CART synthetic data with DP



Confidence when tuning SDC on a map

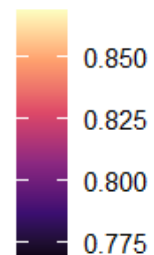
Geographic Uncertainty in Synthetic Data Generation

Expected Disclosure Risk

Average Unique Records over N runs

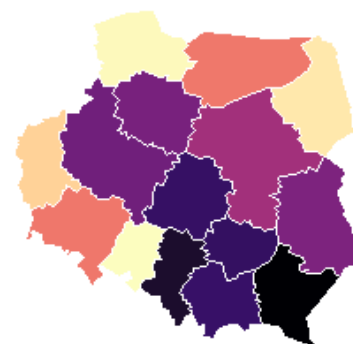


Mean Risk

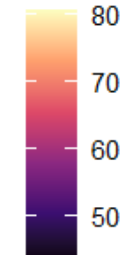


Expected Utility Error

Average Income Dev. over N runs

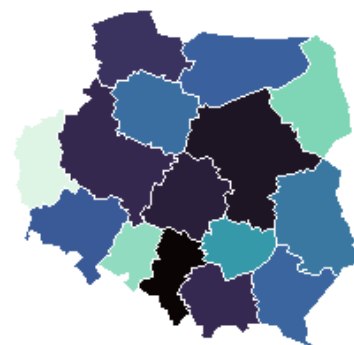


Mean MAE

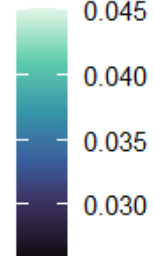


Risk Uncertainty

Volatility of Disclosure Risk

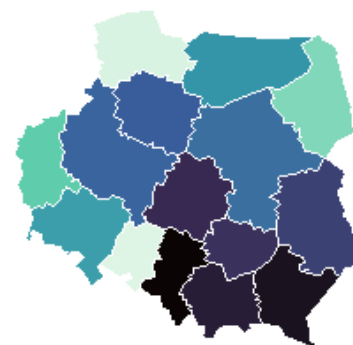


SD of Risk

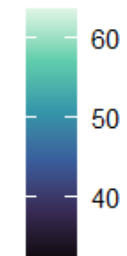


Utility Uncertainty

Volatility of Income Dev.



SD of MAE

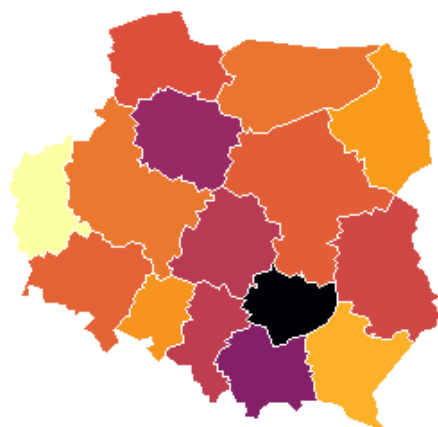


Synthetics with Differential Privacy (DP) / *spatially adapted DP*

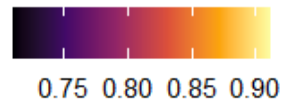
The Spatial Trade-Off of Differential Privacy

Analysis of Poland's 16 Voivodeships using the synthpop SD2011 Dataset

Spatial Risk
(Prop. of Unique Rec.)



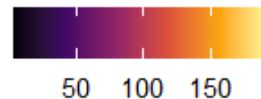
Risk Score



Spatial Utility Error
(Median Income Dev.)



Income Dev.



Privacy Budget: $\epsilon = 3.0$. Notice the visual mirror effect between risk and error.

Thank you!

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Appendix: another uncertainty example

